

Lecture 1 - Big Issues

For four out of the seven big issues mentioned in the lecture (and below), find an app from your own smartphone that you use yourself and describe how this app addresses the respective issue. Please explain your choice in a few sentences each, and add 1-2 explanatory screenshots where needed.

- Limited power supply
 - Rejsekort: Uses a less precise GPS positioning (Approximates it more) to conserve power.
- Limited storage
 - Spotify: Can save lower-quality files of the songs that you download
 - Also uses cache to save storage space when streaming songs, so it doesn't download the whole song.
- Wireless communication channels
 - Snapchat: Can save a video when sending without connection, the video will then be sent whenever you get a proper connection
 - Proportionally scales the quality of pictures and videos depending on your connection strength.
- Limited/different I/O capabilities
 - QR: Uses the camera, which was meant for normal photography for instead inputting data, url, or other text data.
 - Google assistant/Siri/Bixby: Allows for the use of voice to control your phone, example you can say "Hey Google call Alexander" and the phone will then call Alexander.
- Unpredictable usage context
 - Health app: Through an algorithm/gps that can specifically check if you are ACTUALLY moving, or just shaking your phone.
- Privacy & Security
 - Eboks: Uses MitID (2FA) in order to make sure that others can't access your account if they get ahold of your CPR-number.
- Sustainability
 - Device care app: Can be used to optimize power-usage, along with prepping the phone for recycling/refurbishing.